

Study on the Effect of Coal Slit Pore Size on CH₄ and CO₂ Adsorption Characteristics Based on Grand Canonical Monte Carlo

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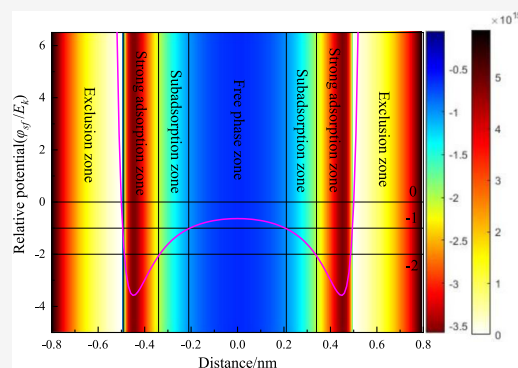
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ABSTRACT: CO₂ Enhanced Coalbed Methane Mining (CO₂-ECBM) technology integrates multiple economic, resource, environmental, and safety benefits such as carbon fixation, enhanced coalbed methane recovery, promotion of natural gas development, and reduction of greenhouse gas emissions. The study found that, based on the adsorption potential theory, the adsorption potentials of CH₄ and CO₂ in the slit pore were calculated, and the pore region was divided based on the potential energy distribution in the pore and the average kinetic energy of gas molecules. The location of positive adsorption potential φ_{sf} was defined as the repulsion region, the region with $|\varphi_{sf}/E_k| > 2$ as the strong adsorption region, the region with $1 < |\varphi_{sf}/E_k| \leq 2$ as the weak adsorption region, and the region with $|\varphi_{sf}/E_k| \leq 1$ as the free-phase region. The present study investigates the adsorption and diffusion characteristics of CH₄ and CO₂ in slit pores of different sizes. This investigation is achieved by conducting Grand Canonical Monte Carlo (GCMC) and Molecular Dynamics (MD) simulations, with the objective of determining the effects of pressure, temperature and slit pore size on the quantity of adsorption were analyzed, and adsorption configurations were obtained under different pressure and pore size conditions. It was found that CH₄ mainly existed in the pores of coal below 3.5 nm, which mainly showed microporous filling and monolayer adsorption. When the slit pore size increased from 2.5 to 8.0 nm, the adsorption isotherm of CO₂ conformed to Langmuir type IV, and the adsorption behaviors included microporous filling, monolayer adsorption and multilayer adsorption, and capillary coalescence phenomenon was observed as the pressure increased. The following essay will seek to demonstrate an understanding of the adsorption-diffusion mechanism of CH₄ and CO₂ in slit pores for improving the recovery rate of CBM, for predicting CBM reserves, and for increasing the potential of CO₂ sequestration.



INTRODUCTION

Coal seam gas as a sustainable energy resource, has been emphasized by countries such as the United States, Canada, Australia and China, and has become the focus of their exploration and development.¹ Data from the International Energy Agency (IEA) show that the reserves of coal seam gas in China are huge and rank among the top in the world.² Meanwhile, the global natural gas demand will grow significantly in the next two decades, and atypical natural gas reservoirs such as shale gas will occupy an increasing proportion.³ The breakthrough of efficient coalbed methane exploration technology is a top priority. As a potent greenhouse gas, CH₄ has a greenhouse effect 20 to 25 times higher than CO₂, hence minimizing CH₄ emissions is vital.⁴ Efficient extraction of coalbed gas is of critical importance for ensuring safety in production in coal mines, optimizing energy structure and environmental protection. Currently, the main extraction technologies include hydraulic stimulation, thermal injection for enhanced product, and gas-assisted production.⁵ CO₂-ECBM technology, integrates many advantages such as carbon capture, augmenting the recovery of coal bed methane

to promote the natural gas industry, and reducing the emission of greenhouse gases, which realizes a win-win situation in terms of economic benefits, resource utilization, environmental protection, and safety guarantee. In China, carbon sequestration technology is one of the core means to achieve the goal of carbon neutrality.⁶

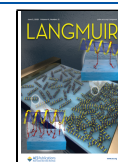
At present, domestic and foreign scholars have conducted a large number of experimental and simulation studies on CO₂-ECBM technology. Sun et al.⁷ used a molecular dynamics simulation to investigate the competitive adsorption behaviors of CO₂ and CH₄, and their mechanisms, at graphene-montmorillonite nonhomogeneous interfaces. The findings revealed that the segmental pressure of the mixed gases is a

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pivotal factor influencing the competitive adsorption, and that this competitive adsorption phenomenon involves multilayer adsorption processes. In another study, Zhao et al.⁸ characterized the pore configuration of 13 coal samples using low-temperature N₂ and CO₂, and found that the characteristics of the pore configuration had a considerable effect on the CH₄ adsorption capacity of coal. The results of the study demonstrated that an increase in mineral content led to a decrease in gas adsorption capacity of coal. Furthermore, the study revealed that the contribution of micropores to the specific surface area of the adsorption pores was more than 99%, which is significantly higher than that of mesopores. Additionally, the study established a strong association between the micropore specific surface area and Langmuir volume. Li et al.⁹ used a Grand Canonical Monte Carlo (GCMC) simulation to investigate the adsorption behaviors of CH₄ and CO₂ in coal at varying moisture levels and pore dimensions, with the objective of comprehending the thermodynamic characteristics of coal adsorption of CH₄ and CO₂. The findings indicated that the adsorption stability and selectivity of the gases diminished with an augmentation of the pore size. Concurrently, the space available for adsorption escalated, culminating in an augmentation of the adsorption quantity. Concurrently, the isosteric heat of adsorption of CH₄ and CO₂ exhibited an initial decrease, followed by an increase, with an increase in water content. Huang et al.¹⁰ examination of the adsorption mechanism of CO₂ and CH₄ in coal seams from a thermodynamic perspective is also of significance. The findings indicated that CO₂ exhibited a greater decrease in Gibbs free energy and entropy during adsorption compared to CH₄, suggesting that the adsorption process of CO₂ was spontaneous. Furthermore, an increase in the quantity of CO₂ resulted in an enhancement of the adsorption of the gas mixture, with CO₂ assuming a predominant role in the process, as evidenced by its higher isosteric heat in comparison to that of CH₄. Zhang et al.¹¹ pioneering development of an in situ ion activation technique enabled the fabrication of ultramicroporous carbon, characterization by a highly narrow pore size distribution. In this process, K⁺ ions were found to distribute uniformly in the chemical bonding of carbon precursors. The experimental data revealed that the adsorption of CO₂ reached 6.26 and 4.17 mmol/g. The CGUC-0.5–6 material demonstrated high selectivity for the binary gas mixtures of CO₂/N₂ and CO₂/CH₄, exhibiting selectivity coefficients of 333.2 and 34.9, respectively. It has also been demonstrated to exhibit both good water vapor resistance and cyclic stability. Wang et al.¹² investigated the consequences of SC–CO₂ on the pore structure of shale under conditions simulating typical reservoir environments. Used XRD, SEM and low-pressure gas adsorption techniques, he conducted a comparative analysis of the alterations in the pore architecture of shale prior to and following SC–CO₂ treatment. The findings revealed that following the SC–CO₂ treatment, there was a decline in carbonate content and an increase in quartz content within the shale. This resulted in substantial alterations to the pore structure. The adsorption selectivity is primarily governed by the pore structure and is found to be inversely correlated with the fractal dimension. Furthermore, it has been established that when the value of the fractal dimension exceeds a certain threshold, the value of the CO₂/CH₄ selectivity factor ($\alpha_{\text{CO}_2/\text{CH}_4}$) declines rapidly ($D_1 > 2.65$, $D_2 > 2.80$).

Qin et al.¹³ investigation set out to examine the adsorption characteristics of shales under varying ratios of CH₄/CO₂. This

was achieved by use a PCTPro-type instrument, which facilitated an in-depth examination of the adsorption behavior exhibited by these shales following supercritical CO₂ treatment. The findings of this study demonstrated that the adsorption capacity for varying CH₄/CO₂ ratios typically decreased post-supercritical CO₂ treatment. Despite the decline, the adsorption selectivity for CH₄/CO₂ still maintained a value greater than 2.5, thereby confirming the robustness of the CS-ESGR adsorption mechanism under these conditions. Chen et al.¹⁴ investigation into the adsorption behavior of graphene-montmorillonite pores for CH₄ and CO₂ in shale slit pores by MD further substantiates these findings. The analysis yielded the conclusion that the heterogeneity of the density arrangement of gases in the slit pores was caused by the nonhomogeneous surface. The adsorption capacity of the graphene exceeded that of the montmorillonite. Furthermore, it was established that CO₂ interacted with the pore walls more strongly than CH₄. The impact of pore size on CH₄ adsorption was found to be more significant, while for CO₂, the effect of the MMT surface was more pronounced.

Scholars at home and abroad have conducted a lot of theoretical research on the conversion of excess adsorption quantity into absolute adsorption quantity. Liu et al.¹⁵ used hybrid grand canonical Monte Carlo/molecular dynamics simulations to study the accessible volumes of He, CH₄, C₂H₆, N₂ and CO₂ gases in a casein matrix and to assess their excess adsorption. It is shown that the accessible volumes obtained from the conventional helium expansion method calculations are overestimated by a factor of nearly two. In the high-pressure phase, the difference between adsorption density and packing density decreases, leading to a reduction in excess adsorption. Helium swelling and helium probes were used to overestimate the accessible volume, and correction factors were introduced for excess adsorption calculations. Babaei et al.¹⁶ used hybrid GCMC and MD for molecular simulation of different slit pore size casein Illite composite shale models at different temperatures, pressures and water contents. It was shown that the flexibility of casein increases the surface roughness and decreases the adsorption density. Water molecules reduce the adsorption sites on the ilmenite surface, and the conversion of excess adsorption to absolute adsorption using adsorption volumes rather than the classical adsorption model has a higher accuracy. Pang and Jin¹⁷ investigated the adsorption behavior of CH₄ in the slit pores of casein from 0.7 to 50 nm using the GCMC method. It was shown that in micropores, excess adsorption has a nonmonotonic relationship with pore size, and when the pore size is larger than 2.5 nm, excess adsorption is independent of pore size. Langmuir, SDR and SBET models fitted using adsorption densities have better performance than fixed densities, and the OK-PSD-L model is presented in the paper to accurately predict the quantity of randomly generated 1250 casein nanoporous media overadsorbed and absolutely adsorbed. The pore size distribution plays an important role in determining the absolute adsorption quantity. Liu et al.¹⁸ calculated the adsorbed phase density by GCMC simulation and used it to calibrate the excess adsorption/desorption isotherm. CH₄ and nC₄H₁₀ adsorbed with increasing pressure at decreasing temperatures, and nC₄H₁₀ had a higher adsorption capacity. There is a hysteresis in the desorption isotherm, which is more significant for nC₄H₁₀ than for CH₄, and the hysteresis decreases with increasing temperature. The adsorbed phase

density is closely related to the system pressure, temperature and pore size. Ghasemzadeh et al.¹⁹ investigated the adsorption density of methane in graphene layers with different pore sizes and temperatures using a hybrid MD/GCMC method, calculated the accessible pore volume using helium swelling and helium probe hair, and discussed the effect of the range of density distribution choices on the calculation of adsorption density. It was shown that it is accurate to use the effective pore width measured by the helium probe as a starting point for calculating the adsorption density. Accurate acquisition of excess adsorption isotherms requires the acquisition of accurate adsorbed phase densities and accessible pore volumes. Liu et al.²⁰ investigated the adsorption behavior of ethane in graphene layers over a range of temperatures, pore sizes and pressures using hybrid GCMC and MD simulations, with ethane and helium probes to measure the accessible volume, and absolute adsorption isotherm analyses based on the Langmuir, Toth and SDR adsorption models. The results show that the adsorption model fits well with the excess adsorption isotherm, but the model cannot accurately predict absolute adsorption. The adsorbed phase density was a function of pressure, temperature and pore size. The adsorbed phase density obtained using density distribution is poor for predicting absolute adsorption, and it is more appropriate to use adsorption volume to estimate absolute adsorption. Jiang and Lin²¹ developed a molecular dynamics model for methane adsorption in nanopores of different widths. It is shown that the absolute adsorption per unit surface area does not increase with the increase of slit pore width when the slit pore diameter exceeds 3 nm, and the Langmuir and Töth equations can be well fitted to the absolute isotherms at different temperatures for a uniform wall surface.

The adsorption potential of CH₄ and CO₂ in slit pores was calculated based on the adsorption potential theory. The pore regions were classified according to potential energy distribution and average kinetic energy of gas molecules. The adsorption behaviors of CH₄ and CO₂ in slit pores of various sizes were studied using GCMC and MD techniques. A detailed analysis was conducted to examine the impact of pressure, temperature, and pore size on the adsorption characteristics. The slit pores were divided into adsorption phase zone and bulk zone by distinguishing the adsorption phase density and gas phase density. This enabled the calculation of absolute adsorption, excess adsorption, and free gas in the pores, which is of great significance for the prediction of coalbed methane reserves. This study makes a significant contribution to our in-depth understanding of the adsorption-diffusion mechanism of CH₄ and CO₂ in the slit pores. This enhanced theoretical understanding is of great value for improving the recovery rate of coalbed methane and the sequestration capacity of CO₂.

EXPERIMENTAL SECTION

Steele 10–4–3 Theory of Adsorption Potential. According to the work of Everett and Powl,²² the potential energy distribution of gas molecules in slit pores was described. The relationship between the potential energy of the gas molecules and the slit pores is governed by the Steele 10–4–3 potential energy function, as illustrated in the following equation (eq 1).

$$\begin{aligned} \varphi_{sf}(D, z) = 4\pi\rho_s\epsilon_{sf}\sigma_{sf}^2\Delta \left\{ \frac{1}{5} \left[\left(\frac{\sigma_{sf}}{D-z} \right)^{10} + \left(\frac{\sigma_{sf}}{D+z} \right)^{10} \right] \right. \\ \left. - \frac{1}{2} \left[\left(\frac{\sigma_{sf}}{D-z} \right)^4 + \left(\frac{\sigma_{sf}}{D+z} \right)^4 \right] \right. \\ \left. - \left[\frac{\sigma_{sf}^4}{6\Delta(D-z+0.61\Delta)^3} + \frac{\sigma_{sf}^4}{6\Delta(D+z+0.61\Delta)^3} \right] \right\} \end{aligned} \quad (1)$$

where φ_{sf} is the adsorption potential between the gas molecules and the slit pore, kJ/mol; ρ_s is the surface density of the atoms on the solid surface of the slit pore, nm⁻²; ϵ_{sf} is the integrated potential energy parameter between the gas molecules and the solid surface of the slit pore, kJ/mol; and σ_{sf} is the collision diameter of the gas molecules interacting with the solid surface of the slit pore. Δ is the interatomic layer distance, nm; z is the perpendicular distance from the gas molecule to the center plane of the slit pore, nm; D is the half-pore width of the slit pore. σ_{sf} and ϵ_{sf} are L-J potential energy parameters, which can be calculated according to the Lorentz–Berthelot mixing rule, as illustrated in the following equation (eqs 2, 3).

$$\sigma_{sf} = (\sigma_{ss} + \sigma_{ff})/2 \quad (2)$$

$$\epsilon_{sf} = \sqrt{\epsilon_{ss} \times \epsilon_{ff}} \quad (3)$$

where σ_{ss} is the collision diameter of atoms on the solid surface, nm; σ_{ff} is the collision diameter of gas molecules, nm; ϵ_{ss} is the potential energy trapping depth (potential well) of atoms on the solid surface, kJ/mol; ϵ_{ff} is the potential energy trapping depth (potential well) of gas molecules, kJ/mol.

Coal is a heterogeneous porous material characterized by intricate physicochemical structures, with carbon atoms forming the core of its macromolecular framework. Other atoms and functional groups are connected to this carbon skeleton through branching or bridging bonds. The carbon skeletons exhibit a microcrystalline structure reminiscent of graphite, and the macromolecular structure of coal features an aromatic ring layered arrangement similar to that of graphite crystals. In this study, the pores within coal are modeled as either slit or cylindrical pores, and the interaction potential energy between gas molecules and the coal walls in the micropores is estimated using the graphite model. According to eq 1, the gas–solid interaction potential for fluid molecules in a slit pore is influenced by both the pore width (D) and the molecular position within the pore (z). Figure 1 provides a schematic representation of the interactions between gas molecules and the walls of the slit pore. The Lennard–Jones potential energy parameters for CH₄, CO₂, and graphite are outlined in Table 1.

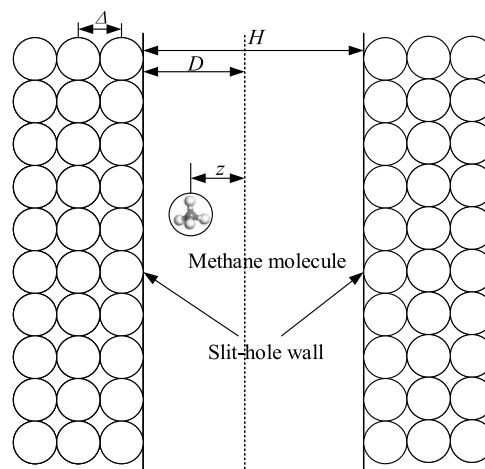


Figure 1. Interaction of gas molecules with the pore wall in the slit pore.

Table 1. L-J Potential Energy Parameters for CH₄, CO₂, and Graphite^{23,24}

molecule type	$\sigma_c(\text{nm})$	$\epsilon/k(\text{K})$
CH ₄	0.373	148.0 ²⁵
CO ₂	0.372	236.1 ²⁶
graphite	0.34	28.0 ²⁶

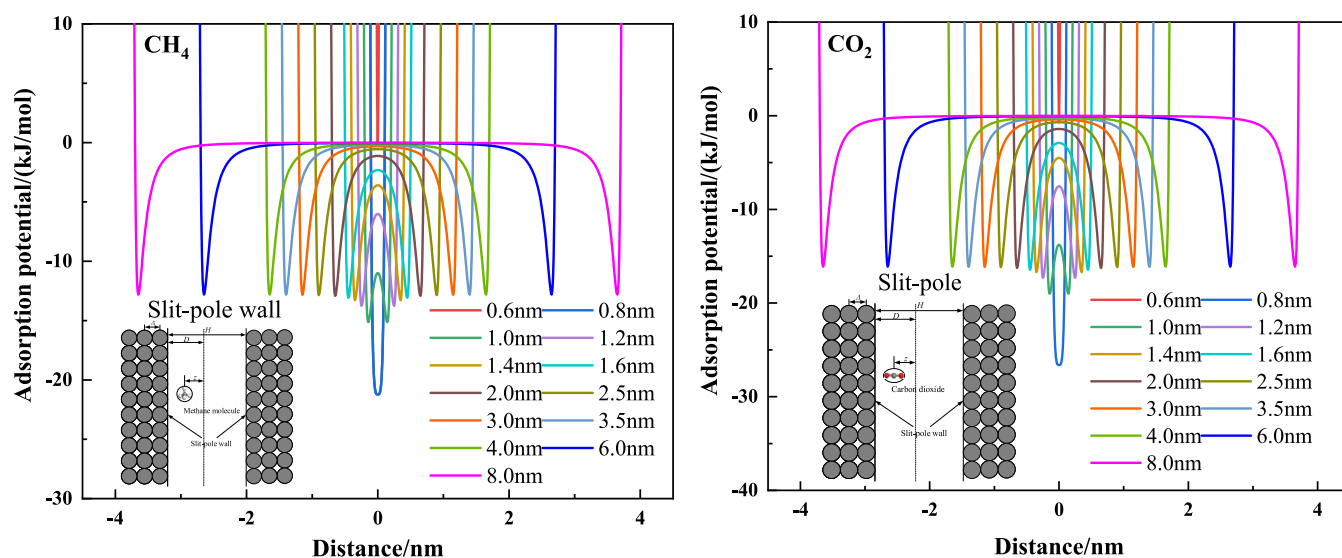
CH₄, CO₂ Adsorption Potential Calculation. Figure 2 depicts the interaction potential energy of CH₄ and CO₂ molecules with the walls of the slit pore. The potential energy curves for both gases exhibit a similar pattern, initially decreasing and subsequently increasing as the distance from the pore wall rises. The CH₄ and CO₂ molecules are subjected to a positive adsorption potential, repulsive force, when they are close to the slit pore wall, and the closer they are to the pore wall, the stronger the positive interaction potential is. As the distance of the molecules from the pore wall increases, the adsorption potential initially decreases sharply, followed by a rapid rise, and eventually increases gradually until approaching zero. Gas molecules in the slit pores exhibit mutual repulsion, then mutual attraction, and finally mutual indifference as the length from the surface increases. In the case of a wider slit pore, the adsorption potential energy distribution map is symmetric centered on the central axis of the pore, showing two lowest potential energy points. However, when the width of the pore decreases, the adsorption potential energies generated by the pore walls begin to superimpose on each other and intensify, resulting in the gradual merging of the two lowest potential energy points, and ultimately forming a single lowest potential energy point on the pore's central axis.

Under specific computational conditions, the interaction potential energy is positive at a pore diameter of 0.6 nm, indicating that the pore wall repels CH₄ and CO₂ molecules, preventing these molecules from entering the pore with a width less than or equal to 0.6 nm. When the pore diameter is below 0.8 nm, there is only one lowest potential energy point, and the lowest potential energy values for CH₄ and CO₂ are −21.222 and −26.614 kJ/mol, respectively. When the pore size exceeds 0.8 nm, the lowest potential energy points become two, which are symmetrically distributed on the central axis. At a pore size of 1.0 nm, the lowest potential energy values for CH₄ and CO₂ decreased to −15.119 and −18.992 kJ/mol, respectively. With the increase of pore diameter, the superposition effect of adsorption potential energy gradually weakened, and the lowest potential energy point first decreased rapidly and then slowly, and when the pore diameter exceeded 1.6 nm, the lowest potential energy value stabilized, indicating that the adsorption capacity of the pore wall

for methane was reduced. The lowest potential energy values of CH₄ and CO₂ were −12.696 and −16.111 kJ/mol, respectively, at a pore size of 8 nm, which decreased by 40.18 and 39.46%, respectively, compared with that of the slit pore of 0.8 nm.

The adsorption potential of coal for CH₄ and CO₂ is related to the physicochemical properties of the gases themselves, and at any slit pore size, CO₂ possesses a greater adsorption potential (interaction) with the coal slit pore wall, resulting in CO₂ being more readily adsorbed by the pore wall compared to CH₄. CO₂ has a linear structure, while CH₄ has an orthotetrahedral structure, and the kinetic diameter of CO₂ is (0.33 nm) smaller than that of CH₄ (0.38 nm), which makes it easier for CO₂ to enter the nanopore space and occupy the dominant adsorption sites. CO₂ has the largest polarizability and quadrupole distance. Highly polarizable molecules are easily polarized in the external electric field and form dipoles, and the quadrupole distance provides an additional electric field attraction that enhances the interaction of the molecules with functional groups on the surface of the solid. CO₂ has a high boiling point, high boiling point gases have strong intermolecular forces, these strong interactions make the gas in the liquid state requires more energy to change to the gaseous state, and therefore in the gaseous state with the solid surface interaction is also stronger. CO₂ possesses a much lower saturation vapor pressure than CH₄, which gives CO₂ a wider adsorption range. In conclusion, the adsorption potential of CO₂ is considered to be higher than that of CH₄.

In this paper, the graphite model is based on the Steele 10–4–3 theory for calculating the adsorption potentials about CH₄ and CO₂. Since graphite is a homogeneous surface material composed of multilayered carbon atoms, with a periodic arrangement of hexagonal lattice structure, a highly homogeneous distribution of electron clouds, and a uniform distribution of electron density in the Fermi surface, which makes the surface adsorption sites highly consistent in terms of the potentials and geometrical distributions. This reduces the preferential aggregation of adsorbates in the local area. For CH₄ and CO₂ molecules, their interactions with the graphite surface are moderate and well matched to the graphite lattice, which makes it easy to achieve uniform coverage. The thickness of the adsorbed layer of CO₂ on the graphite surface is usually larger than that of CH₄. Due to the fact that the adsorption potential of CO₂ is larger than that of CH₄, it has significant physicochemical properties such as quadrupole distance, linear structure, and higher boiling point, which results in the formation of multilayered structure of CO₂ in the high-pressure stage, and the thickness of the adsorbed layer is higher.

**Figure 2.** Illustration of the distribution of adsorption potential for CH₄ and CO₂ gas molecules in slit pores of varying sizes.

Potential Energy Distribution and Region Division. Assuming that the gas molecules are rigid molecules, their average kinetic energy is shown in the following equation (eq 4).²³

$$E_k = 3/2kT \quad (4)$$

where E_k is the kinetic energy of gas molecules, kJ/mol; k is Boltzmann's constant, taken as 1.380649×10^{-23} J/K; T is the temperature of gas molecules, K.

Figure 3 illustrates the distribution of potential energy and region division within the 1.6 nm slit pore. We use the ratio (absolute value)

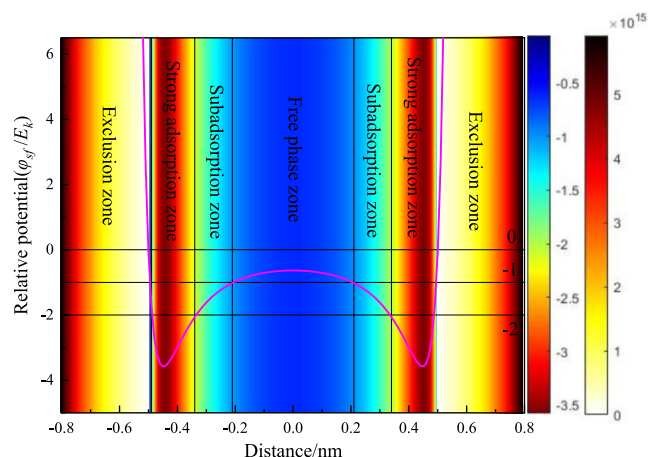


Figure 3. 1.6 nm slit-pore potential energy distribution and region division.

of the interaction potential energy between the pore wall and the gas to the kinetic energy of the molecules as a division criterion to divide the space inside the pore into different regions. When the gas approach the pore wall, the interaction potential energy is positive, indicating the repulsion of the gas molecules, preventing the molecules from entering, this region is known as the repulsion zone, and the closer to the wall, the stronger the repulsion, as shown in the black and red part of the Figure 3.

We define the region where $|\phi_{sf}/E_k| > 2$ as the strong adsorption region, where the potential energy of the pore wall interaction reaches the minimum negative value, which is more than twice of the kinetic energy of the gas molecules, and the gas molecules are strongly attracted to the pore wall in this region, and they can easily gather around the pore wall, and cannot easily be separated from the pore wall due to collisions with other molecules, and it is the region where the gas molecules are the most likely to be adsorbed. The region of $1 < |\phi_{sf}/E_k| \leq 2$ is defined as the weak adsorption region, where the attraction of the pore wall to the gas molecules is relatively weak, and the gas molecules can easily escape from the pore wall due to the collision; The region defined by $|\phi_{sf}/E_k| \leq 1$ is referred to as the free region, where the kinetic energy of the gas molecules surpasses the potential energy of the interaction. In this zone, the attraction between the pore wall and the gas molecules is minimal, allowing the molecules to predominantly exist in a free state. In this region, the kinetic energy of the gas molecules exceeds the interaction potential energy, leading to a weaker attraction between the pore wall and the gas molecules, thus allowing the molecules to predominantly exist in a free state. Gas molecules within the slit pores are only influenced by the wall forces when they are at a certain distance from the wall. The strong adsorption zone and the weak adsorption zone are collectively referred to as the adsorption zone. As the pore diameter increases, the area of the adsorption zone expands; however, the rate of this increase gradually slows down, resulting in a reduced proportion of the adsorption zone. Consequently, the adsorption capacity of the pore for the gas decreases as the pore diameter increases.

Excess Adsorption versus Absolute Adsorption. In the field of supercritical adsorption, Gibbs²⁷ pointed out that the adsorbate

molecules, because of their approximate gaseous density distribution, are independent of the intermolecular forces at the gas/solid interface. Therefore, Gibbs introduced the concept of "excess adsorption", as illustrated in the following equation (eq 5).

$$n_{ex} = \int (\rho(z) - \rho_g) dV \quad (5)$$

where n_{ex} -excess adsorption quantity, mol/g; ρ_g -gas phase density, g/cm³; $\rho(z)$ -adsorbed phase density distribution function normal to the adsorbent surface, g/cm³. The adsorbed phase density distribution function, $\rho(z)$, is difficult to obtain. Gibbs proposes another expression for excess adsorption quantity, as illustrated in the following equation (eq 6).²⁸

$$n_{ex} = n_{ab} - \rho_g V_a \quad (6)$$

where n_{ab} -absolute adsorption quantity, g/cm³; V_a -quantity of adsorbed phase, g/cm³.

The excess adsorption quantity is represented by the absolute adsorption quantity in Figure 4. In the figure, the excess adsorption

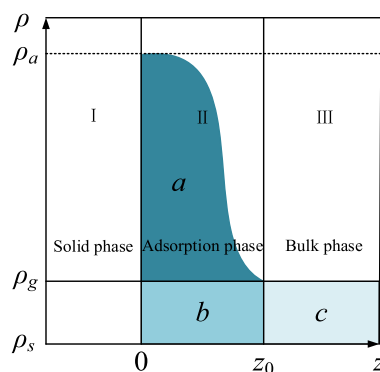


Figure 4. Schematic diagram of excess adsorption versus absolute adsorption.

quantity corresponds to area a , while the absolute adsorption quantity corresponds to area $a+b$. Assuming that the total quantity of adsorbent in the adsorption system is N , this corresponds to area $a+b+c$ in the figure, which can be expressed as $n_{ab} + \rho_g V_g$. The term $\rho_g(V_a + V_g)$ corresponds to area $b+c$ in the figure, and the excess adsorption is shown in the following equation (eq 7).²⁹

$$n_{ex} = N - \rho_g (V_a + V_g) \quad (7)$$

where N -total quantity of adsorbed mass; V_g is the volume of the bulk phase.

The meteorological adsorption density ρ_g can be obtained from the equation of state, while the total quantity of adsorbate N can be determined from the equilibrium equation. $V_a + V_g$ represents the volume of free space in the adsorption system V_p , which is the volume of the system that does not include the adsorbent. This encompasses both the volume of the adsorbed phase and the volume of the bulk phase, and the volume of adsorbed phase and the volume of bulk phase of the adsorption system are constantly changing in the adsorption process. The total gas volume calculated by the MS software is not the whole adsorption gas volume, including the free volume, which needs to be converted into excess adsorption volume from the total gas volume calculated by simulation. According to the isothermal adsorption curve of helium at low pressure by Sorption module, the free volume calculation formula is obtained, and the free volume is calculated as shown in the following equation (eq 8).³⁰

$$V_p = kT \frac{\Delta N^t}{\Delta p} = \frac{RT}{N_A} K \quad (8)$$

where V_p -free volume, cm³; N^t -helium adsorption at pressure p (number of molecules); $k = R/N_A$; N_A is Avogadro's constant, taken as 6.022×10^{23} ; K slope of helium adsorption versus pressure curve.

Table 2. Calculation of Adsorption Phase Density^{1,29}

author	calculation method	note
Arri ³¹	0.425 g/cm ³	equal to the density of liquid methane at room temperature and pressure;
Yee ³²	0.373 g/cm ³	equal to the van der Waals density of methane;
Dubinin ³³	$\rho_a = (8Mp_c)/(RT_c)$	0.372 g/cm ³
Ozawa ³⁴	$\rho_a = \rho_b \exp[-0.0025(T - T_b)]$	ρ_b is the boiling point density of methane at the standardized condition, which is taken as 0.425 g/cm ³ , T_b is the boiling point temperature of methane at atmospheric pressure, which is $T_b = 111.7$ K;
Gensterbluma ³⁵	intercept method to find	extrapolating the excess adsorption and the free phase methane density to satisfy a linear relationship in the Cartesian coordinate system, the intersection with the horizontal coordinate is the adsorbed phase methane density;
Pan ³⁶	0.354 g/cm ³	obtained by the Ono-Kondo modeling;
Li ³⁷	parameter fitting to be determined	meet the adsorption phase methane density value between 0.162–0.425 g/cm ³ .

The CH₄ adsorption phase density was calculated as shown in Table 2, and the accurate acquisition of CH₄ adsorption phase density has an important impact on the calculation of the absolute adsorption quantity. Arri³¹ considered the density of adsorbed phase methane as the density of a CH₄ liquid at room temperature and pressure; In contrast, Yee³² considered the van der Waals density of methane. Dubinin³³ and Ozawa³⁴ proposed empirical equations for the calculation of adsorbed phase density; Gensterbluma³⁵ used the intercept method to find the CH₄ adsorbed phase density; Pan³⁶ calculated the adsorption phase density by Ono-Kondo model. Li³⁷ combined with isothermal adsorption experiments to find the adsorption phase density of CH₄ by the method of coefficient to be determined. In this paper, the adsorption phase density was obtained by the method of coefficient of determination by Li.³⁷

Pressure to Fugacity Conversion. The Peng–Robinson³⁸ equation of state was selected to transform the gas pressure into fugacity as shown in the following equations (eqs 9, 10).

$$P = \frac{RT}{V - b} - \frac{a(T)}{V(V + b) + b(V - b)} \quad (9)$$

$$\ln \frac{f}{P} = Z - 1 - \ln(Z - B) - \frac{A}{2\sqrt{2}B} \ln \left(\frac{Z + 2.414B}{Z - 0.414B} \right) \quad (10)$$

where P is the gas pressure, MPa; f is the gas fugacity, MPa; R is the gas molar constant; V is the gas molar volume, 22.4 L/mol. Z is the compression factor, A and B are the equation coefficients. $a(T)$ is the energy term in the equation of state, Pa·L²/mol².

Thermodynamic parameters of CH₄ and CO₂ are shown in Table 3. Figure 5 illustrates the relationship between the fugacity coefficients

Table 3. Thermodynamic Parameters of CH₄ and CO₂³⁸

absorbent	P_c /MPa	T_c /K	w	molar mass
CH ₄	4.5992	190.56	0.012	16.043
CO ₂	7.3773	304.13	0.224	44.01

and densities of CH₄ and CO₂ as a function of pressure under different conditions. For CH₄, the fugacity coefficient of CH₄ decreases linearly with the increase of pressure, the higher the temperature, the closer the fugacity and the pressure are, and the fugacity of CH₄ decreases with the increase of temperature. The fugacity coefficient of CO₂ decreases linearly with the increase of pressure at 313.15 K. The fugacity exists at an inflection point with the increase of pressure at the other temperatures and the fugacity coefficient decreases abruptly, which is reflected in the density by the existence of a pressure point, which causes a sudden increase of CO₂ density, which is mainly due to the phase transition of CO₂ in the subcritical state. The sudden increase in CO₂ density is mainly due to the phase transition of CO₂ in the subcritical state. The physical properties of CH₄ and CO₂ are detailed in Table 4.

Model Construction. Graphite is a two-dimensional honeycomb crystal structure composed of a single layer of carbon origin with a highly ordered atomic arrangement, and this simple structure is easy to construct theoretical models and can provide a clear theoretical

framework. While coal is an amorphous, nonhomogeneous porous material with complex physicochemical structure, involving the unique nature of coal oxygen-containing functional groups of chemical reactions, porous adsorption properties, the use of Graphite model leads to bias in the results. Therefore, in this paper, a real coal macromolecule model is used for molecular simulation. In this study, a molecular model of gas-fat coal was independently assembled with the molecular formula (C₂₀₇H₁₈₁O₄₇N₃S), and its planar 2D structure is displayed in Figure 6.

The constructed 2D planar model of gas-fat coal was brought into Material Studio software and subjected to hydrogen saturation treatment. The geometry optimization, annealing treatment and dynamics simulation were performed on the model using the Forcite module, and the lowest energy configurations of the gas-fat coal molecules were finally obtained, and the results are demonstrated in Figure 7a. The three-dimensional cell structure of gas-fat coal was created using the Amorphous Cell module, and the six lowest-energy gas-fat coal molecular models were placed into the cell and 3D periodic boundary conditions were introduced. Task selects Construction with a Compass II force field, Quality selects Medium, charge is assigned by Force field assignment, and density is set to 1.14 g/cm³. The temperature is set to 293.15 K. Electrostatic and van der Waals are simulated using “Atom based” approach to simulate the interactions and Lattice type is selected as Cubic. The cell structure of Jixi gas-fat coal was optimized by molecular dynamics method to obtain the minimal energy configuration, as shown in Figure 7b. The optimized cellular structure of Jixi gas-fat coal has a dimension of $A = B = C = 3.12587$ nm, and the density of the cell model is stabilized at 1.14 g/cm³, while the density of the actual measured Jixi gas-fat coal is 1.17 g/cm³, and the simulated value is marginally lower than the measured value. This discrepancy may be caused by the presence of nonmineral substances (e.g., quartz sand, kaolinite) in the coal as well as small-molecule organic matter in the pores that were not modeled, and thus the constructed cell structure is considered reasonable. Further, a series of slit pore models with different sizes were created by incorporating a vacuum layer of varying thicknesses from 0.4 to 8.0 nm between the two gas-fat coal cell structures, as shown in Figure 7c. CH₄ has an regular tetrahedron structure, while CO₂ has a linear triatomic structure. Geometry optimization, annealing treatment and dynamics simulation were performed on these models to obtain the lowest energy configurations of CH₄ and CO₂. After the optimization was completed, the energies of both CH₄ and CO₂ were close to zero.

Simulation Methods and Parameter Settings. The adsorption and diffusion properties of CH₄ and CO₂ in a variety of slit pore models were investigated using GCMC simulation and MD simulation techniques. The Sorption module was used to simulate the adsorption behavior of CH₄ and CO₂ in coal slit pore models of different sizes, with Fixed pressure and Adsorption isotherm selected for the task term, and the Compass II force field was employed. Force field assignment was used to assign charges with a total simulation step of 2×10^7 , of which 1×10^7 was used to equilibrate the frame. Electrostatic was used for Ewald and van der Waals was used for calculating the interactions in an Atom based manner. Where Ewald accuracy was set to 0.001 kcal/mol, Cutoff distance was set to 12.5 Å, and the simulation method was Metropolis. Where Exchange,

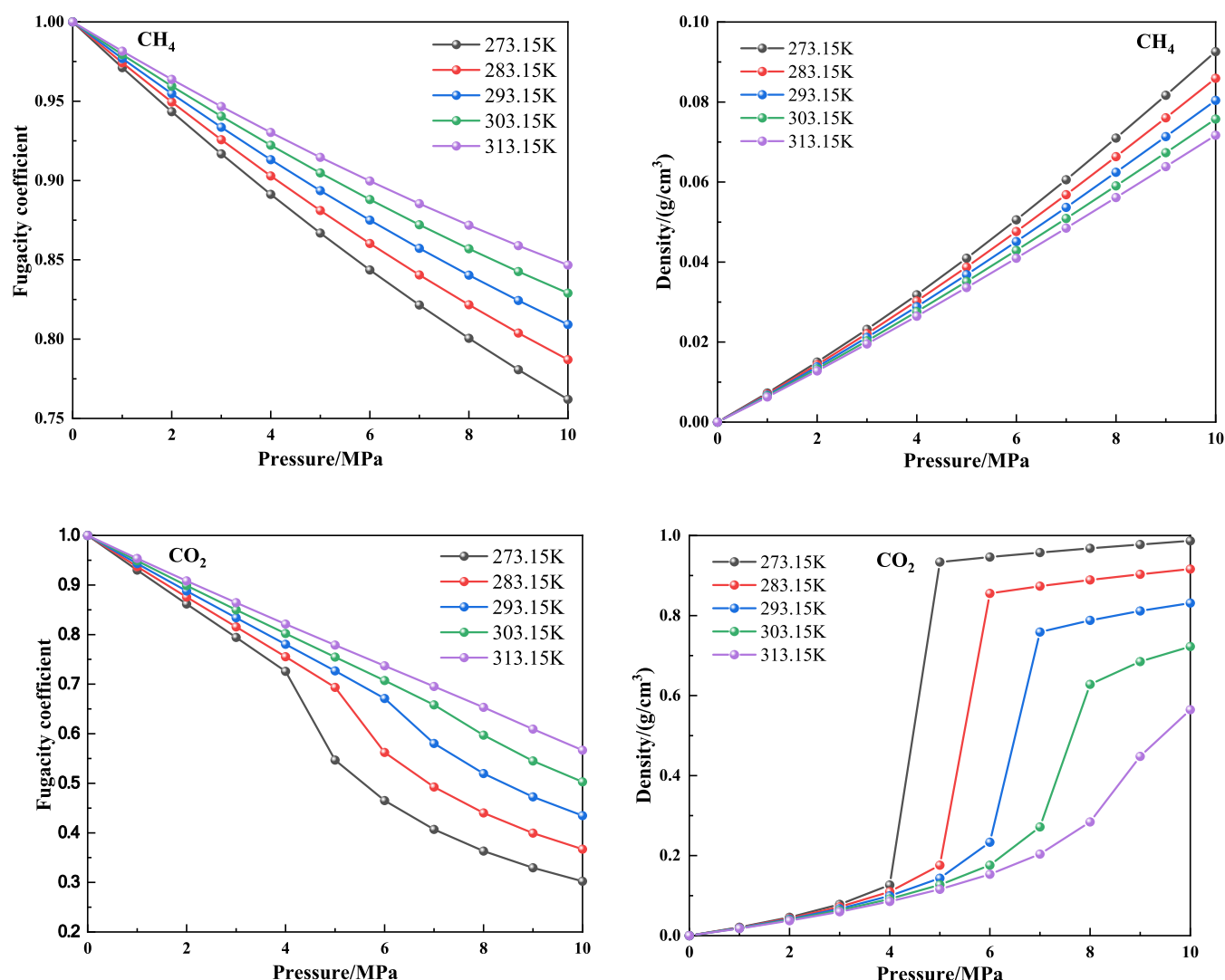


Figure 5. Fugacity coefficients (left figure) and densities (right figure) of CH₄ and CO₂ at different temperatures and pressures.

Table 4. Physical Properties of CH₄ and CO₂³⁸

gas type	geometry	boiling/K	kinetic diameter/nm	polarizability/ 10^{-24} cm ³	dipole moment/ 10^{-18} esu cm	quadrupole moment/ 10^{-26} esu cm ²
CH ₄	regular tetrahedron	111.65	0.38	2.60	0	0
CO ₂	linear	194.65	0.33	2.65	0	4.30

Conformer, Rotate and Translate probabilities are set to 0.39, 0.2, 0.2, 0.2 respectively.

Model Correctness Validation. The experimental and simulated adsorption isotherms of CH₄ by gas-fat coal are shown in Figure 8. To verify the correctness of the GCMC force field parameter settings in this paper, the simulated adsorption data of CH₄ molecules on the gas-fat coal macromolecule structure at a temperature of 298.15 K and a pressure of 0–6 MPa were compared with the experimental data at the same temperature and pressure. It was found that the development trend of the adsorption isotherms between the experimental data and the simulated data was consistent. When the pressure was 6 MPa, the adsorption quantity of CH₄ by gas-fat coal was 15.45 mL/g, the simulation result was 13.17 mL/g, and the absolute error was 2.28 mL/g, which was within an acceptable range. The above proves the rationality of the text's gas-fat coal macromolecule, adsorbate small molecule model and force field parameter settings.

RESULTS AND DISCUSSION

CH₄ Adsorption Quantity. The total gas volume, absolute adsorption quantity, excess adsorption quantity, free gas quantity, and absolute adsorption ratio of CH₄ in coal slit pores with a diameter ranging from 0.4 to 8.0 nm were simulated at a temperature of 293.15 K. The adsorption isotherms at varying temperatures are demonstrated in Figure 9. The simulation results demonstrate the total gas quantity of CH₄ within the slit pores (Figure 9a). This encompasses both adsorbed and free phases. With an augmentation in pore size from 0.4 to 8 nm, the total gas volume of CH₄ in the slit pore escalated from 10.04 to 48.38 mL/g. For slit pores with a pore size of $H \leq 2.0$ nm, the adsorption isotherm of CH₄ conformed to Langmuir type I, while for slit pores with a pore size of $H > 2.0$ nm, the adsorption isotherm of CH₄ exhibited a linear behavior with increasing pressure. In the smaller micropores, the behavior of CH₄ molecules is

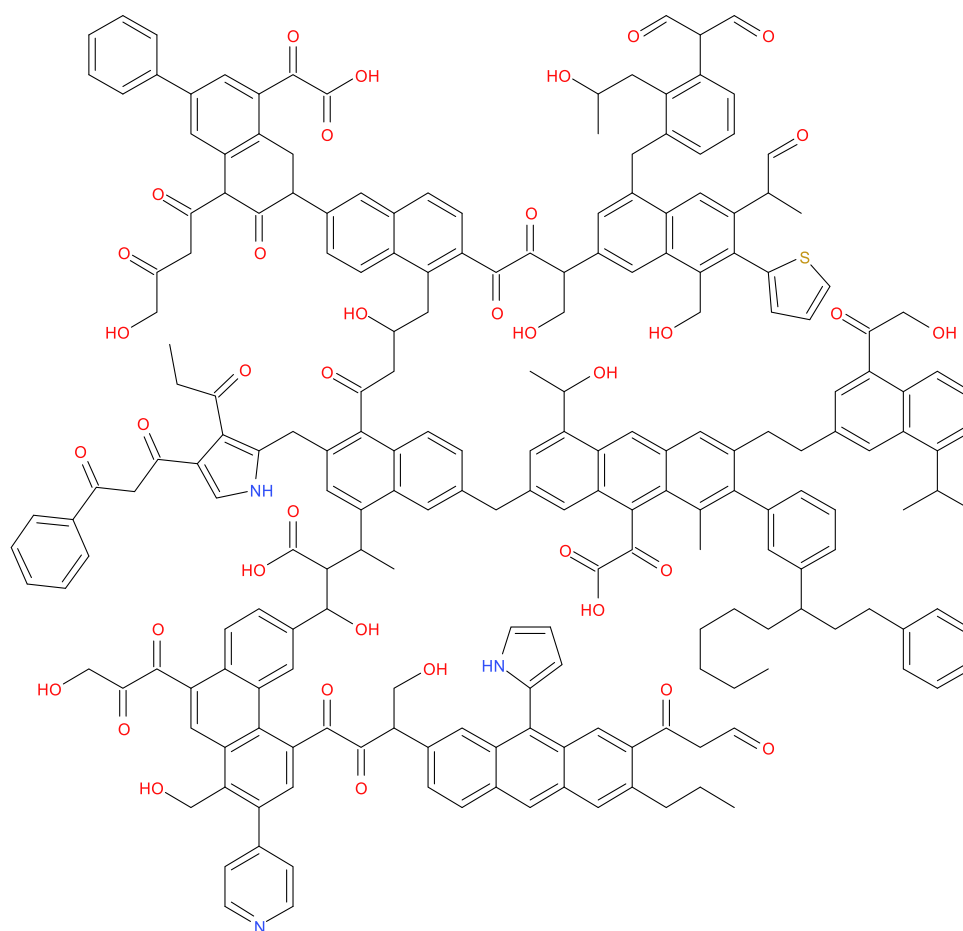


Figure 6. 2D gas-fat coal macromolecular structure model.

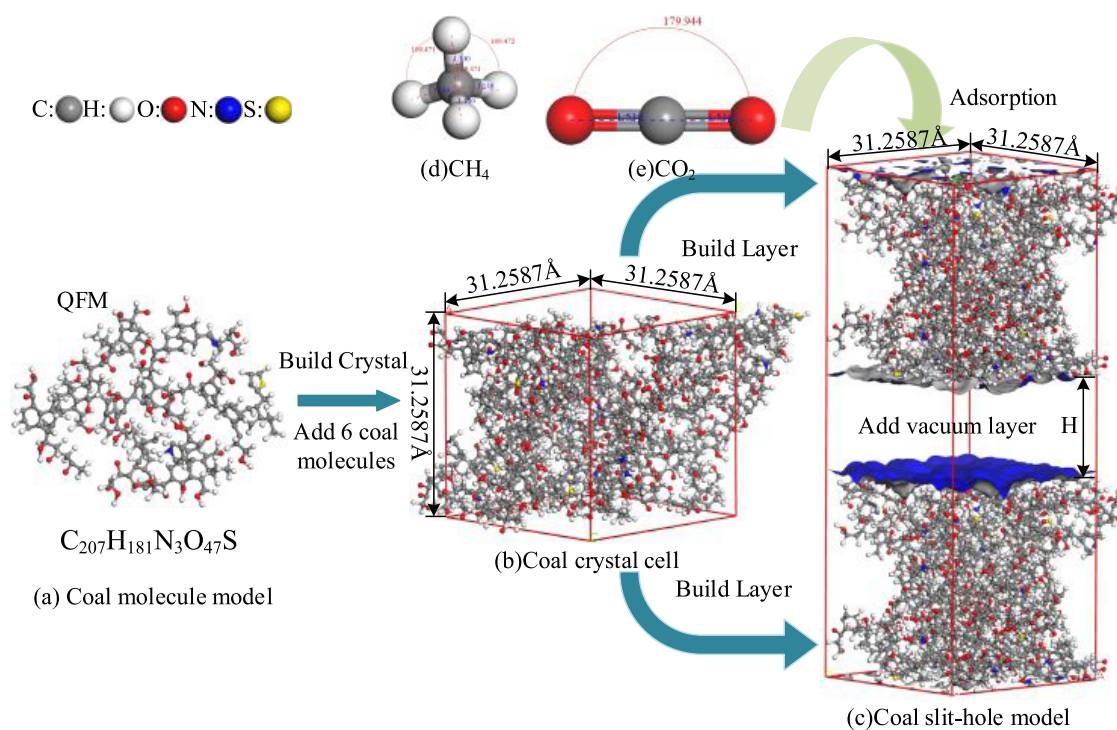


Figure 7. (a–e) CH_4 and CO_2 modeling and coal slit pore construction process.

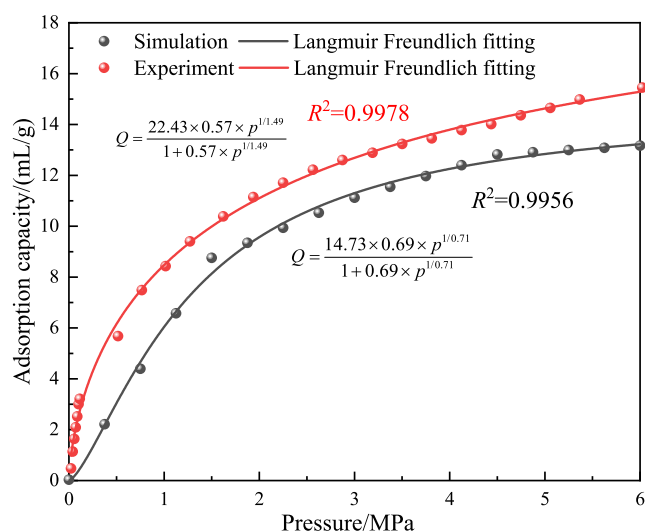


Figure 8. Experimental and simulated adsorption isotherms of CH₄ by gas fertilizer coal.

influenced by the superposition of the adsorption potential field of the surrounding pore wall. This leads to the adsorption behavior of microporous filling, which is limited by the pore volume. Consequently, the growth of adsorption is slow with an increase in pressure, and the adsorption isotherm conforms to the Langmuir I type. In contrast, the surface adsorption behavior in the large-size slit pores is predominantly monolayer adsorption, a phenomenon constrained by the specific surface area. It has been demonstrated that, under pressure, methane molecules possess sufficient kinetic energy to penetrate the pores, resulting in a state that appears to be "adsorbed".

However, it has been determined that methane molecules actually exist in a free state within the large pore structure.

The Figure 9b presents the variation of absolute adsorption of CH₄ in coal slit pores from 0.4 to 8.0 nm under the temperature condition of 293.15 K. At this temperature, the absolute adsorption quantity of CH₄ initially increased with the increase of pore size, reaching a peak of 16.81 mL/g at 3.5 nm, and then decreased with the further increase of pore size, and then the absolute adsorption quantity decreased to 12.97 mL/g when the pore size was 8.0 nm. This finding suggests that the slit pore of 3.5 nm exhibited the optimal adsorption performance for CH₄. Within the low-pressure region ($P \leq 3$ MPa), the slit pores of 0.4 and 0.6 nm exhibited a pronounced adsorption capacity, with the adsorption quantity exceeding that of slit pores of other pore sizes. The Steele10–4–3 potential energy function calculation revealed a potential energy superposition effect in the smaller pores, thereby elevating their adsorption potential energy compared to larger pores. Consequently, under low-pressure conditions, the adsorption rate and quantity were found to be higher in the smaller pores. However, due to the inherent limitations of micropores in terms of volume, the growth of both the adsorption rate and quantity exhibited a gradual decline with an increase in pressure. For slit pores with a pore size greater than 3.5 nm, the adsorption potential field is comparatively low and rapidly decays near the pore wall. The interaction between CH₄ molecules in the region far from the pore wall and the pore wall is weakened, and in some regions, CH₄ molecules are not even affected by the potential field of the pore wall. Despite the availability of sufficient space within these pores to accommodate additional CH₄ molecules, the reduced adsorption potential field results in a decline in their absolute

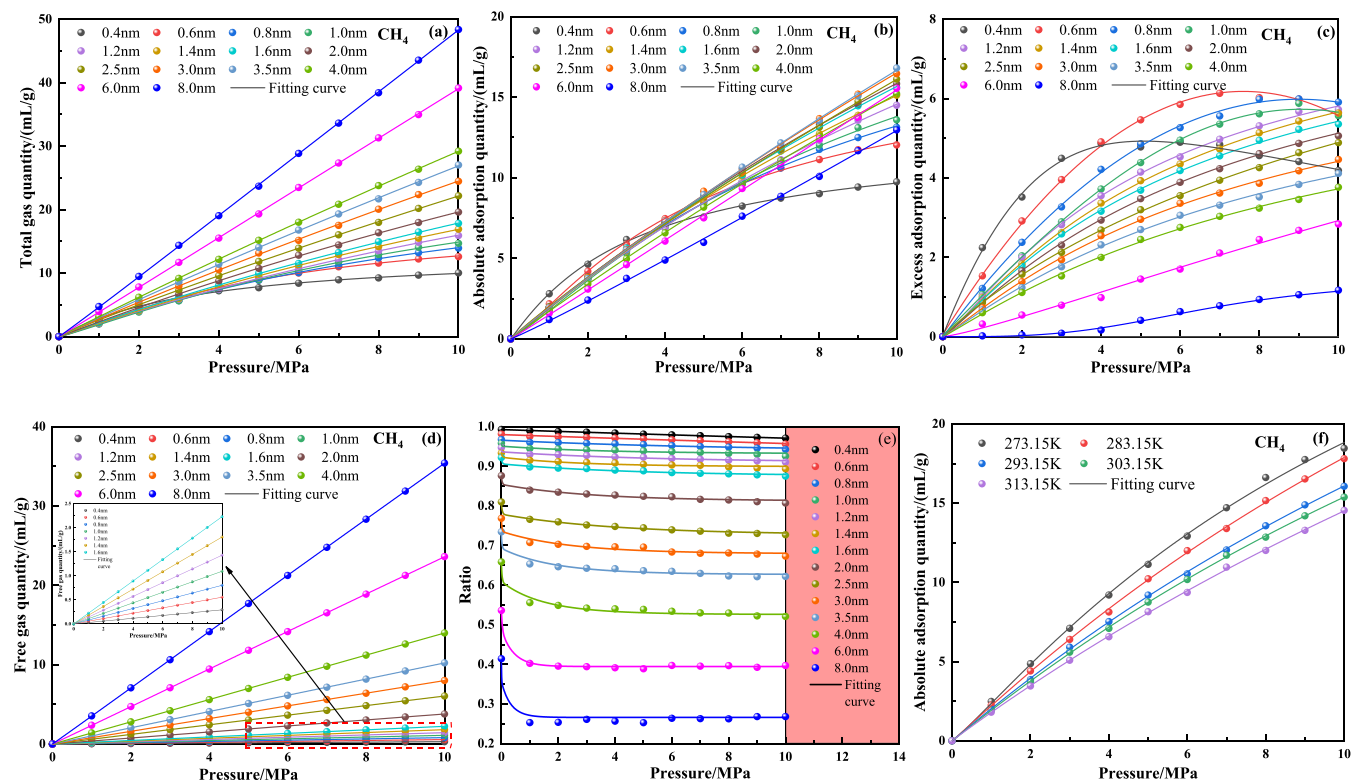


Figure 9. (a–e) CH₄ adsorption isotherms of total gas, absolute adsorption, excess adsorption, free gas, proportion of absolute adsorption, and adsorption at different temperatures in slit pores of different pore sizes.

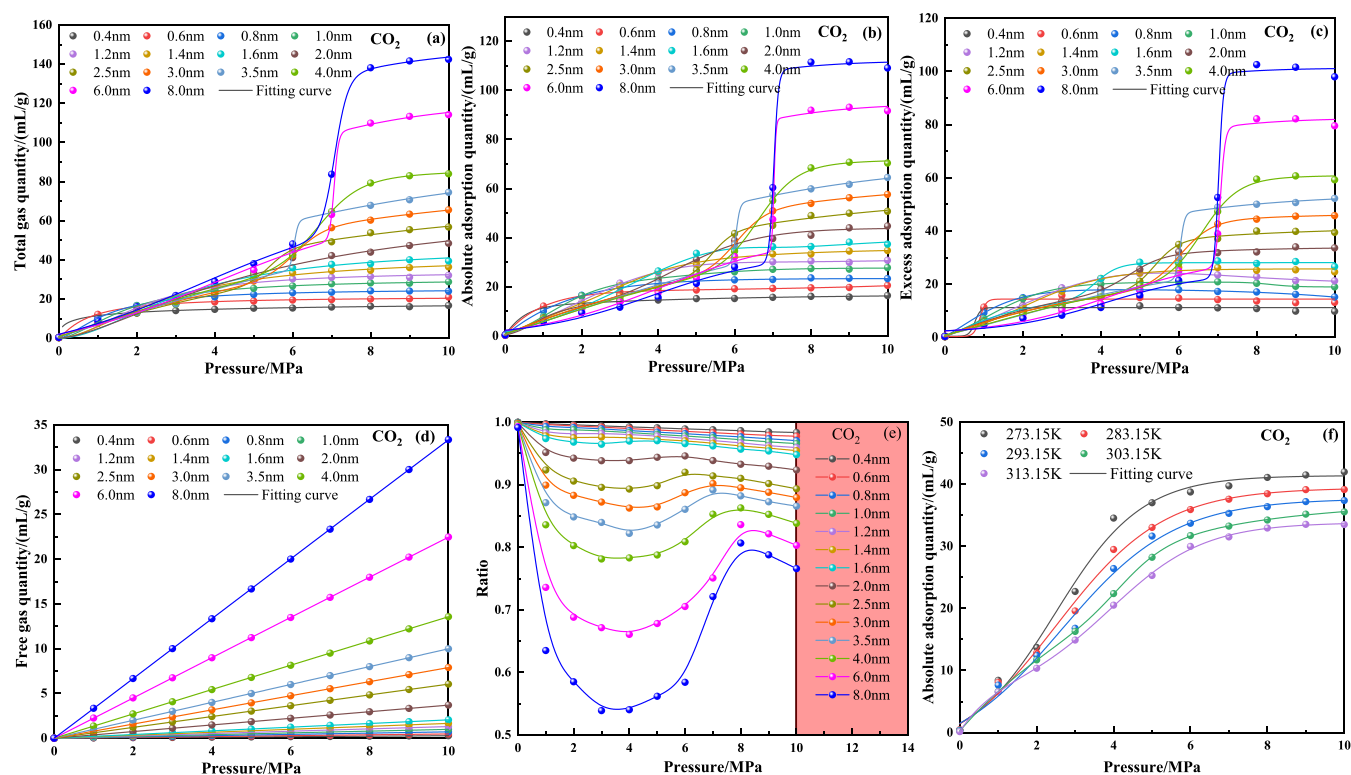


Figure 10. (a–f) Total gas, absolute adsorption, excess adsorption, free gas, proportion of absolute adsorption, and adsorption isotherms at different temperatures for CO₂ in pores of different pore sizes.

adsorption. The predominant existence of CH₄ within coal pores measuring less than 3.5 nm is characterization by two primary endowment forms: microporous filling and monolayer adsorption.

The Figure 9c illustrates the variation of excess adsorption of CH₄ in coal slit pores from 0.4 to 8.0 nm at a temperature of 293.15 K. For slit pores with a pore size of less than 1.0 nm, the excess adsorption of CH₄ increases and then decreases with increasing pressure under constant temperature conditions. Furthermore, the inflection point pressure at which the maximum value is reached increases with increasing pore size. The enhancement of the attraction between the bulk phase interfaces, induced by an increase in pressure, weakens CH₄ adsorption in the vicinity of the solid-phase interfaces, consequently leading to a decrease in adsorption quantity. At pore sizes less than 1.0 nm, the maximum value of excess adsorption initially rises with increasing pore size, reaching 6.13 mL/g at 0.6 nm, and then decreases with further increases in pore size, reaching 5.88 mL/g at 1.0 nm. In the low-pressure region where the pressure $P < 4$ MPa, slit pores with smaller pore sizes demonstrate higher adsorption capacity, which is primarily attributable to the higher adsorption capacity of the small pore size pores due to the potential field. This phenomenon can be attributed to the higher adsorption potential field of the small pore size pores, a consequence of the superposition effect. This finding is consistent with the conclusions drawn by Yin³⁹ and Xiong²³ through molecular dynamics simulations, and by Liu,²⁹ and Monsalvo³⁰ through supercritical isothermal adsorption experiments of CH₄. When the slit pore size is greater than 1.2 nm, the excess adsorption conforms to the Langmuir type I isotherm, which gradually decreases with the increase of slit pore size from 5.72 mL/g at 1.2 nm to 1.17 mL/g at 8.0 nm.

The variation of the free gas volume of CH₄ in the coal slit pores from 0.4 to 8.0 nm under the temperature condition of 293.15 K is demonstrated in Figure 9d. It is evident from the figure that with an increase in pressure, the free gas volume of CH₄ exhibited a linear increase from 0.29 mL/g at 0.4 nm to 35.41 mL/g at 8.0 nm. The initial phase of this increase was characterization by a gradual rise, followed by a subsequent acceleration in the growth rate. This phenomenon can be attributed to the fact that, under increased pressure, the gas molecules acquire sufficient kinetic energy to penetrate the slit pores, akin to a process of "dissolution". Consequently, the CH₄ molecules are adsorbed into the coal body. As the pore diameter expands, the pore volume provided by the slit pores of the coal body gradually increases, which leads to an increase in the quantity of free gas.

As demonstrated in Figure 9e, the proportion of absolute CH₄ adsorbed in coal slit pores decreases with increasing pressure, and the proportion of absolute CH₄ adsorbed decreases with increasing pore size at the same pressure. At low pressure, methane molecules primarily form monolayer adsorption near the pore wall, and methane predominantly exists in the adsorbed state due to the high adsorption potential field of the pore wall. However, with an increase in pressure, methane molecules begin to form monolayer adsorption in the strong adsorption region. This is due to the presence of dispersive repulsion between methane molecules, which results in the newly entering methane molecules being less affected by the adsorption potential. Consequently, these molecules enter the free phase region. In the early stages of adsorption, methane was predominantly in the adsorbed state; however, this proportion gradually decreased with increasing pressure. The average proportion of absolute adsorption decreased from 0.98 at 0.4 nm to 0.27 at

8.0 nm. For slit pores with a diameter smaller than 1.2 nm, the CH₄ was predominantly in the adsorbed state, accounting for over 90% of the total gas volume. Conversely, at pore sizes of 6.0 and 8.0 nm, CH₄ predominantly exists in the free state, accounting for 60.29 and 73.19% of the total gas volume, respectively.

As illustrated in Figure 9f, the absolute adsorption variation of CH₄ in 1.6 nm slit pores is demonstrated at temperatures ranging from 273.15 to 313.15 K. The saturated absolute adsorbed quantity of CH₄ decreased from 18.46 to 14.54 mL/g, representing a 21.24% decrease as the temperature increased from 273.15 to 313.15 K. The increase in temperature is unfavorable to CH₄ adsorption because the increased kinetic energy makes it easier for CH₄ molecules to overcome van der Waals and electrostatic forces and change from the adsorbed state to the free state. This also indicates that CH₄ adsorption in coal is an exothermic process.

CO₂ Adsorption Quantity. As demonstrated in Figure 10a, the isotherms of total CO₂ adsorption in coal slit pores from 0.4 to 8.0 nm at 293.15 K are presented. For slit pores with a pore size of 2.0 nm or less, the total CO₂ adsorption increases rapidly at the beginning and then more gradually as pressure is increased. This is shown as a Langmuir Type I isotherm, and the adsorption at this stage is predominantly microporous filling and is limited by the microporous volume. As the pore size increases from 0.4 to 2.0 nm, the total adsorbed quantity of CO₂ increases from 16.70 to 48.45 mL/g. The process of CO₂ adsorption was found to be enhanced at low pressures, where the pore size *H* increased from 2.5 to 8.0 nm, encompassing microporous filling, monolayer adsorption and multilayer adsorption. With the rise in pressure, an inflection point was observed in the adsorption isotherm, accompanied by a surge in the adsorption quantity, suggesting that CO₂ began to undergo capillary coalescence within the pores. Following the attainment of a state of saturation, the further increase in pressure resulted solely in adsorption on the outer surface, and the adsorption quantity stable, as evidenced by the manifestation of a Langmuir type IV isotherm. The total gas volume of CO₂ increased from 56.77 to 114.15 mL/g as the pore size increased from 2.5 to 8.0 nm. The onset of capillary coalescence was observed at 5 MPa for a pore size of 2.5 nm, while coalescence occurred at 7 MPa for a pore size of 8.0 nm, indicating an increase in coalescence pressure with increasing pore size. In the high-pressure region, CO₂ undergoes capillary coalescence, and the gas transforms into liquid before reaching saturation, a phenomenon that is governed by the pore size.

As demonstrated in the study by Rafati,⁴⁰ capillary coalescence of CO₂ transpires at minimum porosities of 2.5 nm. This phenomenon transpires under the prerequisite that the slit pores are sufficiently capacious to permit accommodation of a minimum of four layers of adsorbate molecules; it does not occur in pores which are only capable of adsorbing a quantity of no more than two layers.

Kelvin⁴¹ posited the theory that the saturation vapor pressure of a liquid is contingent upon the curvature of the playing surface. In eq 11, Kelvin expressed the relationship between the saturation vapor pressure *p* of a droplet of radius ρ and the saturation vapor pressure *p*₀ of a flat liquid surface.

$$\ln(p/p_0) = -(2\gamma V_m / rRT) \cos \theta \quad (11)$$

where *V*_m is the molar volume of the liquid, γ is the surface tension of the liquid, *R* is the gas constant, 8.314 J/(mol·K); *T*

is the absolute temperature, K; *r* is the radius of the capillary tube, nm; ρ is the radius of curvature of the liquid surface of the bending moon, nm; θ is the contact angle of the liquid with the wall of the capillary tube.

As demonstrated in eq 11, the saturated vapor pressure of the liquid within the capillary tube is lower than that of a flat liquid surface, resulting in the liquid surface within the capillary tube rising and the occurrence of capillary coalescence. This phenomenon is illustrated in Figure 11. The increase in

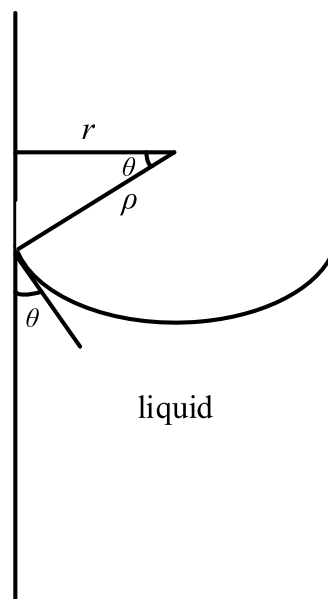


Figure 11. Curved moon liquid level in a capillary tube.

pressure during capillary coalescence is observed to be directly proportional to the aperture of the slit, thereby validating the simulation results presented in this study.

As demonstrated in Figure 10b, the variation of absolute adsorption of CO₂ in coal slit pores from 0.4 to 8.0 nm at 293.15 K is illustrated. It is evident that as the pore diameter increased from 0.4 to 2.0 nm, the absolute adsorption of CO₂ increased from 16.43 to 44.75 mL/g. This observation is consistent with the adsorption behavior at this time conforming to the Langmuir type I isotherm. Subsequently, when the pore size was augmented from 2.5 to 8.0 nm, the absolute adsorption quantity of CO₂ increased from 50.73 to 109.09 mL/g, a phenomenon attributed to the occurrence of capillary coalescence. This resulted in the transformation of the adsorption isotherm into a Langmuir type IV.

The Figure 10c illustrates the excess adsorption of CO₂ in the 0.4 to 8.0 nm coal slit pores at 293.15 K. It is evident that the absolute adsorption of CO₂ increased with increasing pore size. In the 0.4 and 0.6 nm coal slit pores, a rapid increase in excess adsorption to saturation was observed when the pressure reached approximately 1 MPa, and subsequently, there was a negligible change with pressure, demonstrating typical microporous filling behavior. Furthermore, an increase in excess adsorption from 15.18 to 24.61 mL/g was observed with an increase in pore size from 0.8 to 1.4 nm. The adsorption isotherms were found to be in accordance with Langmuir Type I for pore sizes less than 1.4 nm and Langmuir Type IV for larger pore sizes (i.e., 1.6 nm and above). The quantity of free gas (CO₂) in the coal slit pores was found to increase linearly (i.e., in a straight line) with increasing

pressure from 0.27 mL/g at 0.4 nm to 33.35 mL/g at 8.0 nm (see Figure 10d).

Figure 10e provides a visual representation of the adsorption of CO₂ to the total gas volume ratio, as well as the effect of pressure on different coal slit pores. For pore sizes below 2.5 nm, the ratio of absolute adsorbed quantity decreased with increasing pressure. This ratio also decreased when the size of the pores increased, e.g., it decreased from 98.35% at 0.4 nm to 92.37% at 2.0 nm at 10 MPa pressure. Conversely, for pore sizes of 2.5 nm and above, the proportion decreased initially, subsequently increased, and then decreased again with increasing pressure. The enhancement of the pore size has been observed to result in a weakening of the stacking effect and a decline in the adsorption potential field at the pore wall, consequently leading to a reduction in the ratio of adsorbed phases. At the phase transition point of pressure, the phenomenon of capillary coalescence occurs, resulting in a surge in the quantity of adsorbed gas. Consequently, the ratio of adsorbed phase gas experiences an increase. Beyond a certain threshold of pressure, the pore reaches a state of saturation with adsorbed CO₂, leading to a plateau phase in the total adsorbed gas volume. This results in a decline in the adsorbed gas ratio.

As illustrated in Figure 10f, the absolute adsorption of CO₂ in the 1.6 nm slit pore exhibited a temperature-dependent behavior, ranging from 273.15 to 313.15 K. It is evident from the data presented that the absolute adsorption quantity of CO₂ decreased from 41.99 to 33.51 mL/g as the temperature was increased from 273.15 to 313.15 K. The molecular dynamics theory posits that the aggregation of gas molecules on a solid surface is contingent on the number of colliding molecules and the residence time. At a constant pressure, an increase in the absolute temperature of the system has been shown to both increase surface tension of the pore wall and the magnitude of Brownian motion of the gas molecules. The consequence of this increase in motion is an enhancement of both the adsorption and desorption rates. However, it is important to note that, since adsorption is an exothermic process and desorption is an absorptive process, the desorption rate tends to increase more significantly. This increase in desorption rate leads to the desorption process being favored, resulting in the adsorbed molecules detaching from the solid surface and becoming free molecules. In conclusion, this decrease in the absolute adsorption quantity can be attributed to this desorption process.

The phenomenon of coal bed gas adsorption is primarily observed to occur on solid surfaces. The adsorption potential on the pore walls is identified as the predominant driving force of the adsorption process. As demonstrated schematically, the adsorption potential of CO₂ is greater than that of CH₄, with 2.39 times more CO₂ adsorbed than CH₄ in a slit pore of 1.6 nm, providing theoretical support for the CO₂-ECBM (enhanced coke breeze-making) technology. The kinetic diameter (D_{kin}) of adsorbed molecules dictates the effective pore range into which they can penetrate. With a value of 0.33 nm, the D_{kin} of the adsorbed CO₂ molecules is slightly less than that of CH₄ molecules (0.38 nm). The smaller molecular kinetic diameters of CH₄ allow it to enter smaller micropores, which typically have stronger adsorption potentials. This results in greater CO₂ adsorption potential. Evaporation is a process that is more likely to occur for CH₄ due to its lower boiling point when compared with that of CO₂. The presence of a large number of acidic functional groups on the coal

surface, which include oxygen and nitrogen, can produce strong electron-transfer interactions with CO₂. The latter has a permanent quadrupole moment and high polarizability, which means that it is more susceptible to polar molecules than nonpolar ones. This results in induced dipoles. These induced dipoles, in turn, interact with intrinsic dipoles on the coal surface, thereby generating induced forces that augment electrostatic interactions between the coal surface and the CO₂ molecules.

CONCLUSIONS

The region with positive values of the interaction potential φ_{sf} is referred to as the repulsion region, the region with $|\varphi_{\text{sf}}/E_{\text{k}}| > 2$ is referred to as the strong adsorption region, the region with $1 < |\varphi_{\text{sf}}/E_{\text{k}}| \leq 2$ is referred to as the weak adsorption region, and the region with $|\varphi_{\text{sf}}/E_{\text{k}}| \leq 1$ is referred to as the free-phase region. In smaller micropores, CH₄ adsorption is mainly manifested as microporous filling, which is limited by the pore volume. While in the larger slit pores, CH₄ adsorption mainly shows monolayer adsorption, limited by the specific surface area. CH₄ mainly exists in the pores below 3.5 nm of the coal, in the form of microporous filling and monolayer adsorption. For the slit pores with pore size less than 1.2 nm, CH₄ mainly exists in adsorption state, accounting for more than 90% of the total gas volume. And when the pore size reaches 6.0 and 8.0 nm, CH₄ exists mainly in the free state, accounting for 60.29 and 73.19% of the total gas volume. For CO₂, its total adsorption in the slit pore increased from 16.70 to 48.45 mL/g when the pore size increased from 0.4 to 2.0 nm, which was in accordance with Langmuir type I isotherm. When the pore size increased from 2.5 to 8.0 nm, its total gas volume of CO₂ increased from 56.77 to 114.15 mL/g, which conformed to the Langmuir type IV isotherm, and the adsorption behaviors included micropore-filling, monolayer adsorption and multi-layer adsorption, and capillary coalescence phenomenon was observed with the increase of pressure.

ASSOCIATED CONTENT

Data Availability Statement

The data sets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author Contributions

J.J.: Methodology, Supervision, Conceptualization, Funding acquisition. Y.X.: Conceptualization, Investigation, Writing-original draft, Writing-review and editing. All authors have read and agreed to the published version of the manuscript.

Notes

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